Network geometry and phase structure

The correlation network of the 26 business cycles, summarized by the mean edge length of its minimum spanning tree (distance $\sqrt{2(1-\rho)}$, 10-year windows), and the phase lead of each economy against the world cycle. The network has tightened over five decades but does not contract before crises, and the economies are too tightly phase-locked for an epicenter to be recovered.

Status key

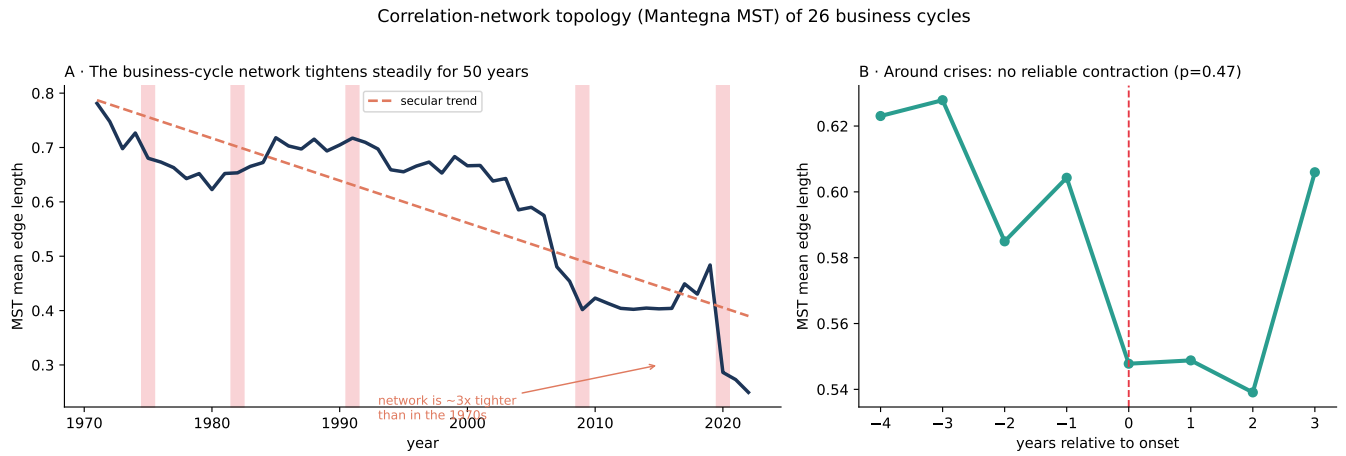
CURRENT	Paper data (Maddison 2023) and the paper's conventions; numbers match docs/methods.md.
NEEDS RELABELLING	Numbers right, but an annotation on the figure is out of date (said in the caption).
CONVENTION W	Hilbert transform applied to 1970–2022 only (convention W); the paper uses P.
SUPERSEDED	Earlier data or crisis definition; the record of how the paper got here.
DESCRIPTIVE	Atlas material from the Global Macro Database; describes, does not test.
UNVERIFIED SERIES	The series drawn matches no stored results file; redraw before use.

In this book

F5.1	Correlation-network topology: the minimum spanning tree of 26 business cycles	NEEDS RELABELLING
F5.2	Companion version: the network has tightened about twice since the 1980s	CURRENT
F5.3	Pairwise phase lags: is there an epicenter?	CONVENTION W

F5.1 NEEDS RELABELLING

Correlation-network topology: the minimum spanning tree of 26 business cycles

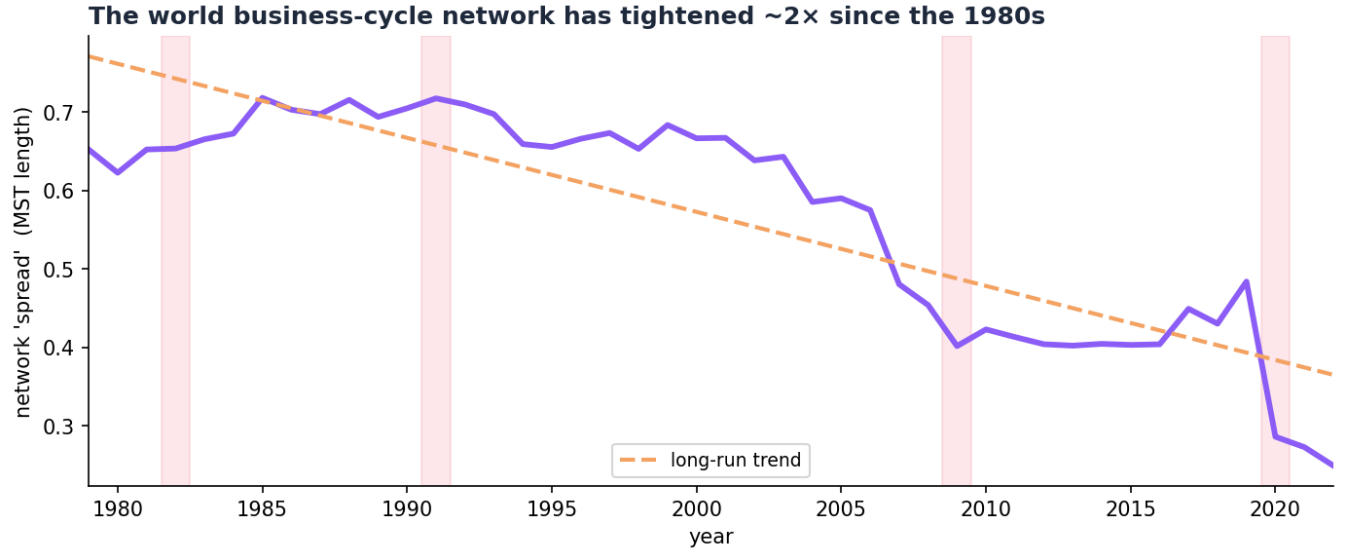


(A) Mean MST edge length $L(t)$, 10-year windows, with a linear trend: the network tightens for 50 years. (B) L around crisis onsets. Two annotations are out of date. '~3× tighter' compares the noisy single-year endpoints (0.781 in 1971, 0.249 in 2022); decade means give 1.9× (0.706 in the 1970s, 0.379 in 2013–22). 'p = 0.47' could not be reproduced; the stored test is $\tau = -0.171$, p = 0.167 (raw) and $\tau = -0.086$, p = 0.316 (detrended). Both say no pre-crisis contraction.

Numbers Most central economies (mean MST degree): France 2.85, Netherlands 2.65, Finland 2.62.
Made by paper1/scripts/eraA_claude_ai/mst_network.py → mst_results.json (reproduces exactly) · **Data** Maddison 2023
File figures/05_network_phase/F5-01_mst_network.pdf

F5.2 CURRENT

Companion version: the network has tightened about twice since the 1980s

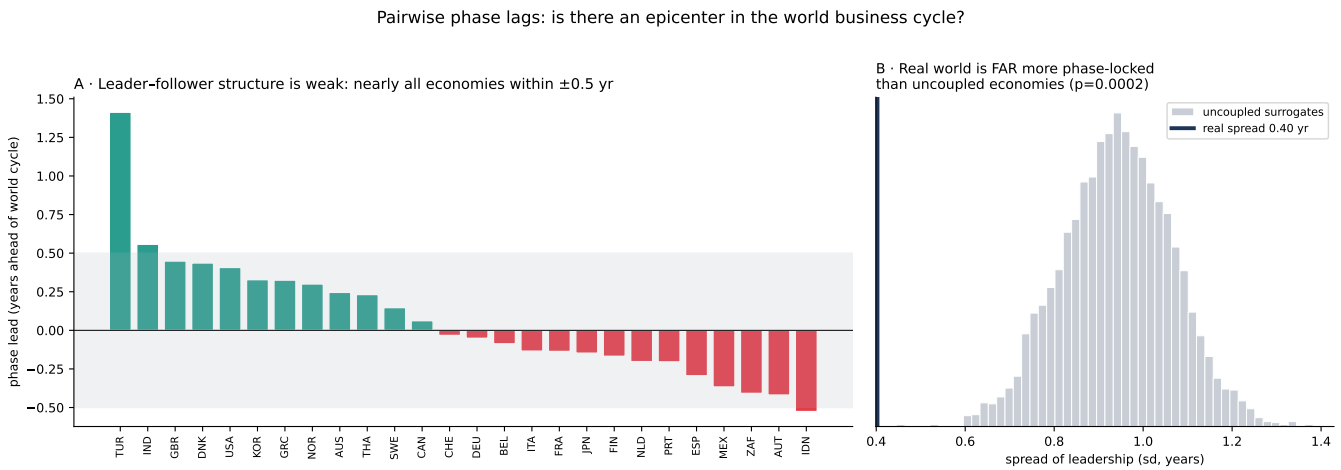


$L(t)$ from 1979 with its long-run trend. With windows kept inside 1970–2022 the decade-mean comparison is 0.652 (1979–89) against 0.379 (2013–22), about 1.7×; '~2×' is a fair round figure (Methods §7.7).

Numbers 1.7× (1979–89 base) to 1.9× (1970s base).
Made by Companion-book figure A9 · **Data** Maddison 2023
File figures/05_network_phase/F5-02_companion_network_tightening.png

F5.3 CONVENTION W

Pairwise phase lags: is there an epicenter?



(A) Each economy's mean phase lead over the world cycle in years (mean period 5.85 years). Nearly all sit within ±0.5 years; Turkey is the outlier at +1.4. (B) The spread of leads (s.d. 0.40 years) against uncoupled surrogates (mean 0.94, minimum 0.45). The economies are phase-locked far more tightly than chance, so no leader can be recovered. The figure's p = 0.0002 is the correct lower-tail value; phase_lag_results.json stores p_spread = 1.0, which is the upper tail. Computed with convention W (phases from 1970–2022 only), although Methods §1 states P for every phase; Methods §6.6 quotes this W value. Recompute under P.

Numbers Spread 0.403 years; lower-tail p = 0.0002 (all 5,000 surrogates larger).
Made by paper1/scripts/eraA_claude_ai/phase_lags.py → phase_lag_results.json (reproduces exactly) · **Data** Maddison 2023
File figures/05_network_phase/F5-03_phase_lags.pdf